

Akila Abenayaka

Kegalle, Sri Lanka | akilaabenayaka27@gmail.com | +94 71 4604907
[GitHub](#) | [LinkedIn](#) | [Medium](#) | [Portfolio](#)

SOFTWARE ENGINEERING UNDERGRADUATE

Passionate Software Engineering undergraduate at the University of Kelaniya with strong interests in **Generative AI**, **Full-Stack Development**, and **Cloud-Native Systems**. Experienced in developing scalable web applications using MERN and Spring Boot ecosystems while exploring AI-powered solutions using LLMs and RAG architectures. A fast learner and collaborative team player committed to building impactful, intelligent software solutions.

TECHNICAL SKILLS

Languages	: Java, JavaScript, TypeScript, Python, C, HTML, CSS, SQL
Frontend	: React.js, Next.js, Redux Toolkit, Tailwind CSS, Bootstrap
Backend	: Node.js, Express.js, Spring Boot, REST APIs
AI & Data	: Generative AI, Gemini API, RAG Architecture, Prompt Engineering
Databases	: PostgreSQL, MongoDB, MySQL, Redis
Tools & Platforms	: Git, Docker, Firebase, Supabase, Postman, Vercel

EDUCATION

BSc (Hons) Software Engineering <i>University of Kelaniya</i>	2022 – Present <i>Sri Lanka</i>
---	------------------------------------

- Specializing in **Net-Centric Web Application Development**, **Data Science & Engineering**, and **Mobile Computing**
- CGPA: **3.729 / 4.0**

GCE Advanced Level <i>Kegalu Vidyalyaya</i>	2007 – 2020 <i>Kegalle, Sri Lanka</i>
---	--

- Completed coursework in **Combined Mathematics**, **Physics**, and **Chemistry**
- 2A & 1C Passes** | Z-Score: **1.589**

PROJECTS

AI-Driven Intelligent Chatbot <i>Individual Project</i>	<i>Python, Gemini 1.5 Pro, PostgreSQL</i>	GitHub
--	---	------------------------

- Developed an AI-powered chatbot utilizing **Gemini 2.5 Pro** and **Retrieval-Augmented Generation (RAG)** for context-aware responses
- Built a custom knowledge retrieval system using **PostgreSQL** for efficient document querying and low-latency responses
- Integrated vector search and PDF querying capabilities to enhance external knowledge access
- Optimized prompt engineering techniques to improve conversational quality and response accuracy

Rabbit E-Commerce Platform <i>Individual Project</i>	<i>MERN Stack, Redux Toolkit, Stripe</i>	GitHub
---	--	------------------------

- Engineered a scalable full-stack clothing marketplace with secure authentication and responsive UI design
- Managed application-wide state using **Redux Toolkit** to ensure seamless frontend data flow
- Integrated **Stripe** for secure payment processing and **Cloudinary** for media management
- Implemented RESTful APIs and optimized frontend responsiveness across mobile and desktop devices

- Developed a vendor reservation and management system for BMICH book fair operations
- Built secure backend services using **Spring Boot**, **JWT**, and **Spring Security**
- Designed scalable architecture patterns to improve maintainability and booking performance
- Created professional dashboards for vendor and inventory management workflows

- Contributed to the official TEDx UOK event website with emphasis on UI/UX and mobile responsiveness
- Integrated **Supabase** services for secure event registration and real-time backend functionality
- Improved website SEO and optimized performance for faster asset loading and user engagement

CERTIFICATIONS & ACHIEVEMENTS

- **First Place** – Paramount Mathz IQ Competition, Department of Mathematics, University of Kelaniya
- **Generative AI Fundamentals** – Google Cloud Training (2024)
- **Java (Basic/Intermediate) Certification** – HackerRank (2023)

VOLUNTEER EXPERIENCE & LEADERSHIP

Head of Logistics
Software Engineering Students' Association (SESA)

2025 – Present
University of Kelaniya

- Coordinating logistics and operational planning for technical events and workshops
- Managing resource allocation and vendor coordination for student activities

Digital Marketing Lead
IEEE Student Branch

2024 – 2026
University of Kelaniya

- Leading digital outreach campaigns and social media engagement for IEEE events
- Designed promotional content and improved audience reach for technical webinars

LANGUAGES

English

: Professional Proficiency

Sinhala

: Native Proficiency

REFERENCES

- **Dr. Nalin Warnajith**
Head, Software Engineering Teaching Unit, University of Kelaniya
nwarnajith@gmail.com | +94 71 7570140
- **Dr. Isuru Hewapathirana**
Senior Lecturer (Grade II), Software Engineering Teaching Unit, University of Kelaniya
ihewapirana@kln.ac.lk | +94 777 329626